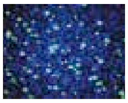
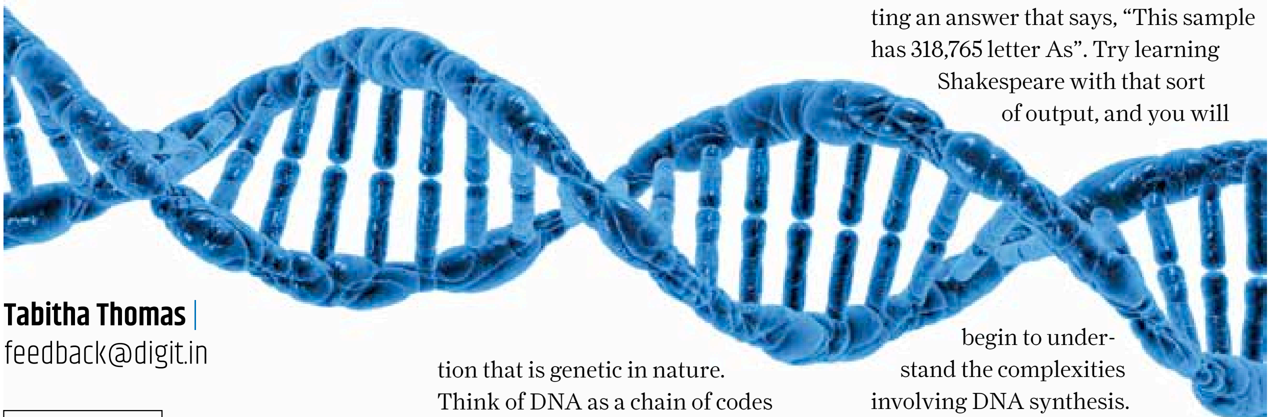


Eating poop-grown microbes
A new study says that microbes might be able to feed on poop and their extracted components could be used to grow edible bacteria. **http://dgit.in/PoopFeast**

Orange cave crocodiles
The crocodiles in the caves of Gabon feasting on bats and moving in guano might be evolving into a new species. **http://dgit.in/GabonCroc**

SINGLE CELL DNA SEQUENCING

Every single cell in your body could potentially hold the cure for cancer. Single cell sequencing can revolutionise the field of medical science and research and save thousands of lives



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What do you think when we say DNA? If you are a Bones fan, you are probably picturing lab researchers working with strands of hair on a jacket or saliva on a postal stamp trying to catch a bad guy. However, there is so much more to DNA than you can begin to imagine.

WHAT IS DNA?

To better understand single cell DNA sequencing, let's go back to the basics. DNA or deoxyribonucleic acid is a material present in each and every living organism. This self replicating material is also a carrier of all information that is genetic in nature. Think of DNA as a chain of codes that are put together in a very specific order that is unchangeable and thus works as a marker for your lineage and heritage. All human beings start their life cycle as an embryo. Within this single celled embryo lie genome markers that will determine the type of fully formed human it will grow into. Now, 99% of the material found in DNA across humans is the same. It's the 1% that makes us look remarkably different from each other.

DNA SEQUENCING?

There are DNA molecules and then there is sequencing of those molecules. The laboratory method to determine this exact sequence is called DNA Sequencing. Think of it as coming across a library of alien books. You could quickly understand their alphabet by noting down each individual character. If it was English that was the alien language, for example, you would know all of the letters from A to Z, even if you didn't know the order. But that's just the first step. What we've done so far in sequencing DNA is break the library down into an alphabet, and find the general length of the average book, and some choice phrases. Single cell sequencing is like trying to nail down the exact sequence of letters in each book in that library. It's hard, because you have to imagine that the books are microscopic in size, and the smallest sample you are able to test usually have dozens of books. Then, you add in the complexity of not being able to read the letters directly, but instead have to use chemicals. So it's like throwing a chemical in with the sample and getting an answer that says, "This sample has 318,765 letter As". Try learning Shakespeare with that sort of output, and you will begin to understand the complexities involving DNA synthesis.

WHY IS IT POPULAR?

For years now, the studies conducted by the fields of omics (neologism in the English language that refers to biological study fields that end with -omics) like genomics, metabolomics, proteomics etc. have clubbed together a culture of tissues or cells assuming (incorrectly) that all cells in a single organism (or organ) are alike. This 'clubbing together' means that the heterogeneity or individuality, if you will, of the single cell is lost.

In the world of biology, heterogeneity can be explained at three levels:

(i) different organisms are heterogeneous